

Computer Organization & Bus Structures

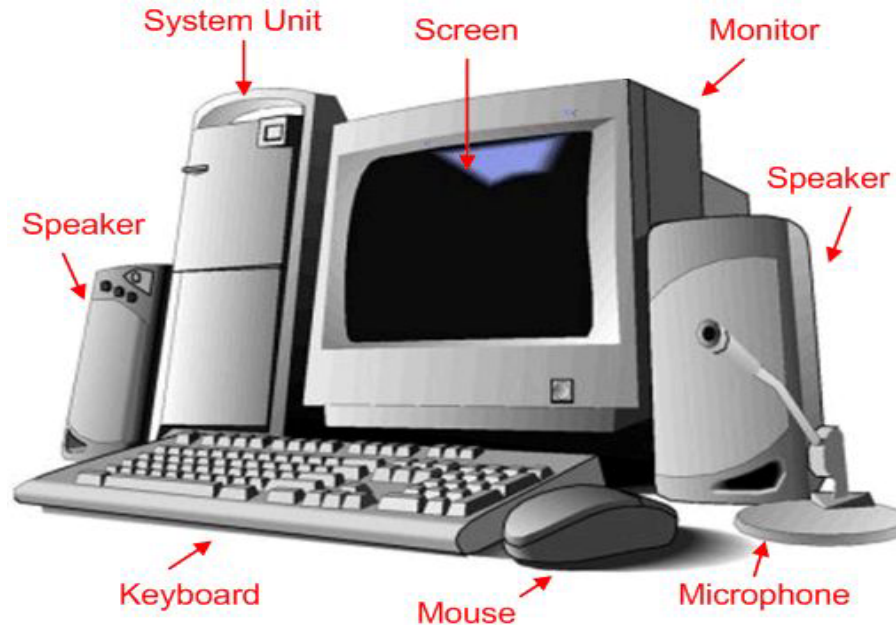
Microprocessor:

Microprocessor is a multipurpose programmable logic device that reads binary instruction from a storage device memory and accept binary data as input and processes the instruction for execution.



Microcomputer:

- A microcomputer is a complete computer on a smaller scale and is generally a synonym for the more common term, personal computer or PC , a computer designed for an individual. A microcomputer contains a microprocessor (a central processing unit on a microchip), memory in the form of read-only memory and random access memory , I/O ports and a bus or system of interconnecting wires, housed in a unit that is usually called a motherboard .



Brief History of the Computers:

- 1946 The first generation of Computer ENIAC was started to be used based on the vacuum tube technology.
- 1958 the first transistorized computer TRADIC was announced by IBM.
- 1959 first IC was invented.
- 1960s ICs were started to be used in CPU boards.
- 1970s entire CPU was put in a single chip. (1971 the first microprocessor of Intel 4004 (4-bit data bus and 2300 transistors))
- Late 1970s Intel 8080/85 appeared with 8-bit data bus and 16-bit address bus and used from traffic light controllers to homemade computers.
- 1981 First PC was introduced by IBM with Intel 8088 microprocessor.
- Motorola emerged with 6800. Apple Macintosh computers started to use 68000 series of microprocessors.

Evolution of Intel 80x86 Family Microprocessors:

Processor	Year Intro.	Transistors	Clock Rate (MHz.)	External Data Bus	Internal Data Bus	Add. Bus
4004	1971	2,250	0.108	4	8	12
8008	1972	3,500	0.200	8	8	14
8080	1974	6,000	3	8	8	16
8085	1976	6,000	6	8	8	16
8086	1978	29,000	10	16	16	20
8088	1979	29,000	10	8	16	20
80286	1982	134,000	12.5	16	16	25
80386DX	1985	275,000	33	32	32	32
80386SX	1988	275,000	33	16	32	24
Pentium C	1993	3,100,000	66 -200	64	32	32
Pentium MMX	1997	4,500,000	300	64	32	32
Pentium Pro	1995	5,500,000	200	64	32	36
Pentium II	1997	7,500,000	233-450	64	32	36
Pentium III	1999	9,500,000	550-733	64	32	36
Itanium	2001	30,000,000	800-...	128	64	64

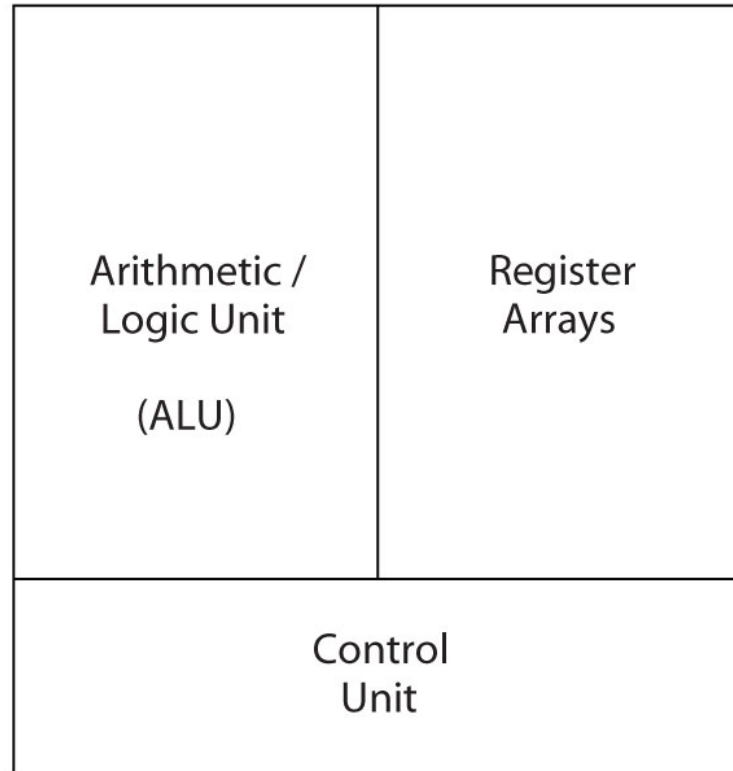
Microcomputers and Microprocessors:

There are three major parts of a Computer System.

1. Central Processing Unit (CPU): Also simply called as the microprocessor acts as the brain coordinating all activities within a computer.
2. The Memory: The program instructions and data are primarily stored.
3. The Input/ Output (I/O) Devices: Allow the computer to input information for processing and then output the results. I/O Devices are also known as computer peripherals.

The integrated Circuit (IC) chip containing the CPU is called the microprocessor and the entire computer including the microprocessor, memory and I/O is called a microcomputer.

Microprocessor (MPU):

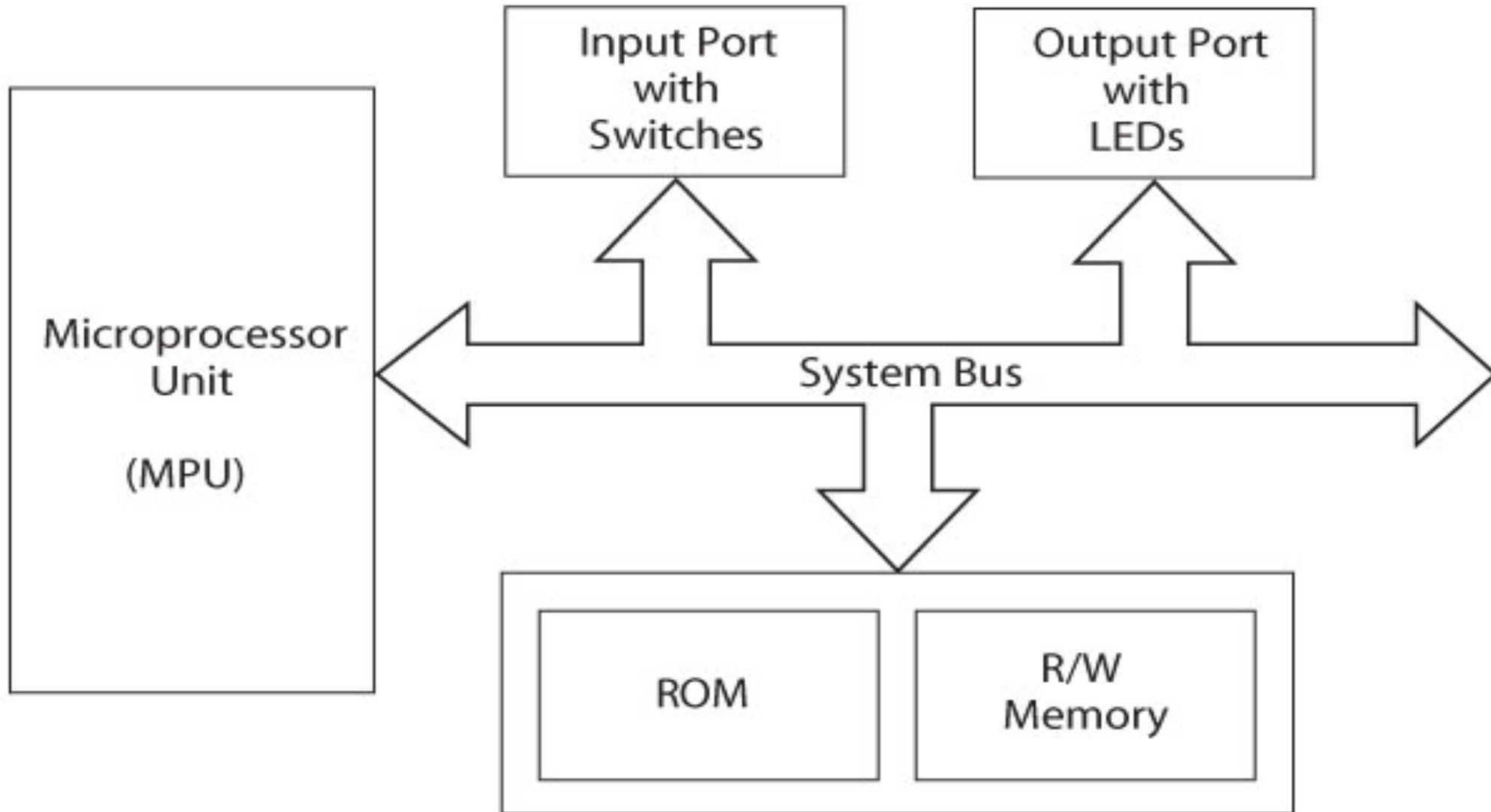


- MPU (CPU)
 - Read instructions
 - Process binary data

Microprocessor Architecture:

- MPU communicates with Memory and I/O using the System Bus
 - Address bus
 - Unidirectional
 - Memory and I/O Addresses
 - Data bus
 - Bidirectional
 - Transfers Binary Data and Instructions
 - Control lines
 - Read and Write timing signals

Microprocessor-Based Systems:

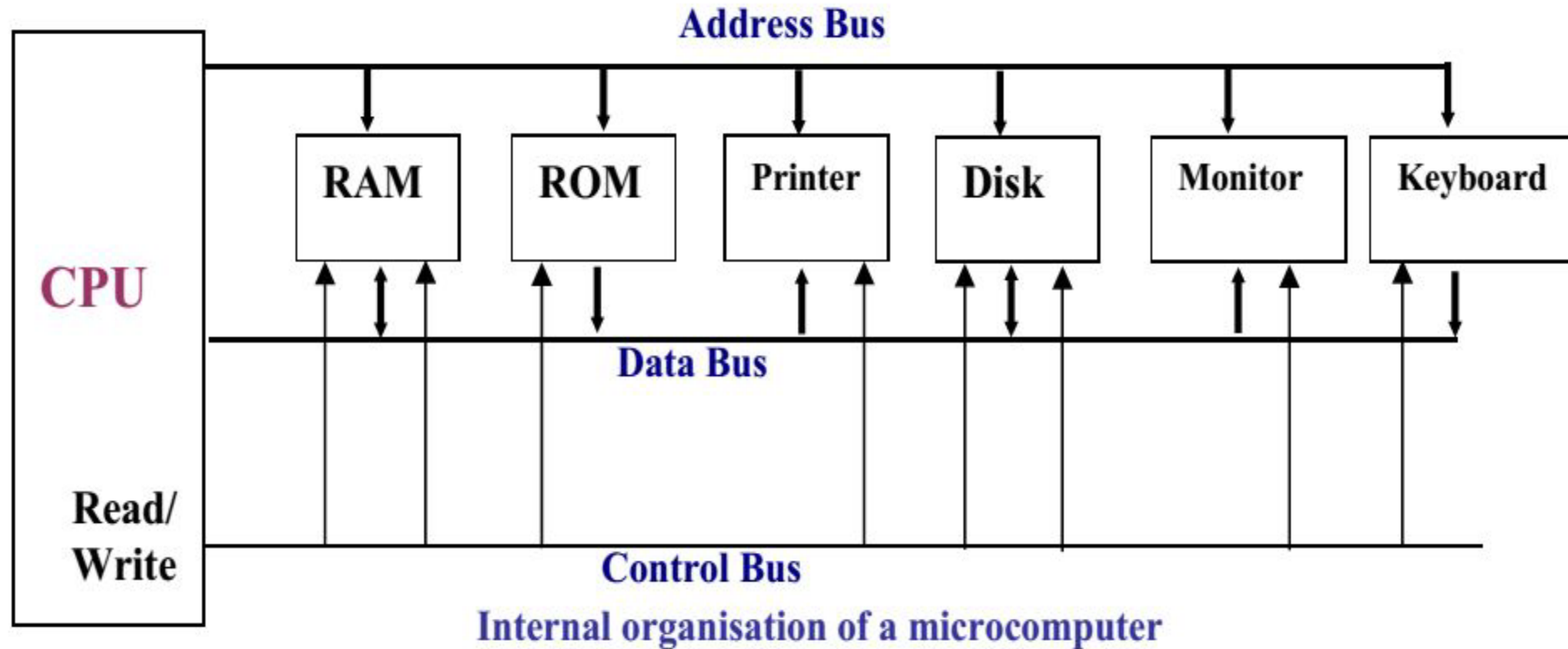


Bus organization:

The CPU is connected to memory and I/O devices through a strip of wires called a bus. The bus inside a computer carries information from place to place. In every computer there are three types of busses:

1. **Address Bus:** The address bus is used to identify the memory location or I/O device the processor intends to communicate with. The width of the Address Bus ranges from 20 bits (8086) to 36 bits for (Pentium II).
2. **Data Bus:** Data bus is used by the CPU to get data from / to send data to the memory or the I/O devices. The width of a microprocessor is used to classify the microprocessor. The size of data bus of Intel microprocessors vary between 8-bit (8085) to 64-bit (Pentium).
3. **Control Bus.** How can we tell if the address on the bus is memory address or an I/O device address? This where the control bus comes in. Each time the processor outputs an address it also activates one of the four control bus signals: Memory Read, Memory Write, I/O Read and I/O Write.

Internal Organization of Microcomputer:



Microcomputer Memory:

- The address and control bus contains output lines only, therefore it is unidirectional, but the data bus is bidirectional.
- There two types of memory used in microcomputers:
 - **RAM(Random Access Memory/ Read-Write memory)**is used by the computer for the temporary storage of the programs that is running. Data is lost when the computer is turned off. So known as Volatile memory.
 - **ROM(Read Only Memory)** the information in ROM is permanent and not lost when the power is turned off. Therefore, it is called nonvolatile memory.

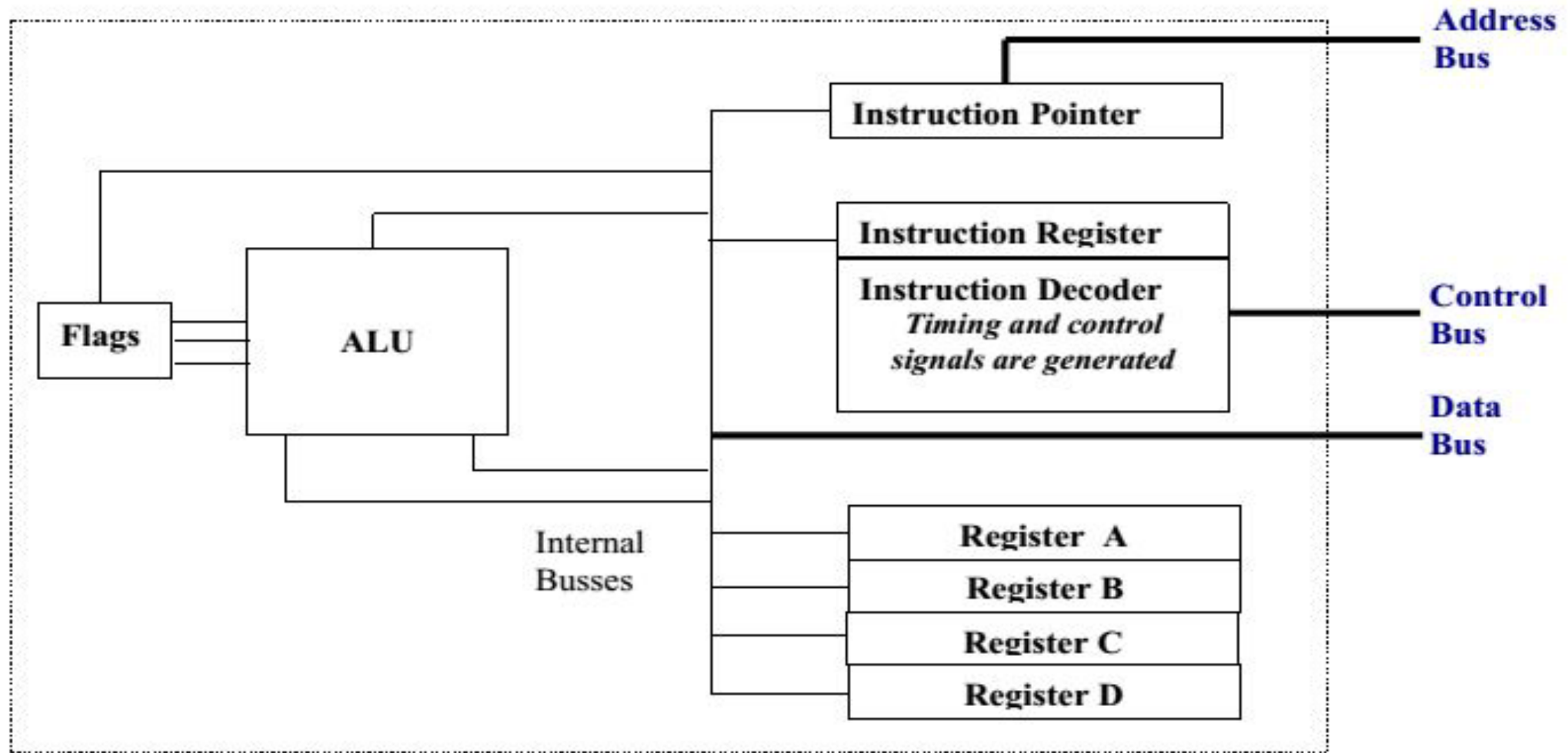
Note that RAM is sometimes referred as primary storage, where magnetic /optical disks are called secondary storage.

2. Memory

- physical devices used to store data or programs (sequences of instructions) on a temporary or permanent basis for use in an electronic digital computer.
- Computer main memory comes in two principal varieties: [random-access memory](#) (RAM) and [read-only memory](#) (ROM).
- RAM can be read and written to anytime the CPU commands it, but ROM is pre-loaded with data and software that never changes, so the CPU can only read from it.
- ROM is typically used to store the computer's initial start-up instructions.
- In general, the contents of RAM are erased when the power to the computer is turned off, but ROM retains its data indefinitely.
- In a PC, the ROM contains a specialized program called the [BIOS](#) that orchestrates loading the computer's [operating system](#) from the hard disk drive into RAM whenever the computer is turned on or reset.

Inside the CPU:

- A program stored in the memory provides instructions to the CPU to perform a specific action. This action can be a simple addition. It is function of the CPU to fetch the program instructions from the memory and execute them.
 1. The CPU contains a number of registers to store information inside the CPU temporarily. Registers inside the CPU can be 8-bit, 16-bit, 32-bit or even 64-bit depending on the CPU.
 2. The CPU also contains Arithmetic and Logic Unit (ALU). The ALU performs arithmetic (add, subtract, multiply, divide) and logic (AND, OR, NOT) functions.
 3. The CPU contains a program counter also known as the Instruction Pointer to point the address of the next instruction to be executed.
 4. Instruction Decoder is a kind of dictionary which is used to interpret the meaning of the instruction fetched into the CPU. Appropriate control signals are generated according to the meaning of the instruction.



Internal block diagram of a CPU

3. I/O Unit

- **Input/output (I/O)**, refers to the communication between an [information processing system](#) (such as a [computer](#)), and the outside world possibly a human, or another information processing system.
- [Inputs](#) are the signals or data received by the system, and [outputs](#) are the signals or data sent from it
- Devices that provide input or output to the computer are called [peripherals](#)
- On a typical [personal computer](#), peripherals include input devices like the keyboard and [mouse](#), and output devices such as the [display](#) and [printer](#). [Hard disk drives](#), [floppy disk drives](#) and [optical disc drives](#) serve as both input and output devices. [Computer networking](#) is another form of I/O.

What is an Embedded System?

Definition:

- An embedded system is a combination of hardware and software designed to perform a specific dedicated function.

Examples:

- Washing machines
- Smart watches
- Traffic light controllers
- Industrial robots

Embedded System Hardware Overview:

Main Hardware Components:

1. Processor (Microcontroller or Microprocessor)
2. Memory (ROM, RAM, Flash)
3. Input/Output Interfaces
4. Timers and Counters
5. Communication Interfaces (UART, SPI, I2C, etc.)
6. Power Supply

Processor(CPU):

- The brain of the embedded system.
- Executes instructions and controls all operations.

Types:

- Microcontroller (MCU): CPU + Memory + I/O on a single chip (e.g., Arduino, PIC, ARM Cortex-M)
- Microprocessor: Needs external memory and I/O (e.g., Raspberry Pi, Intel 8086)

Memory:

❑ Used for storing program and data:

- ROM: Stores firmware permanently
- RAM: Temporary data storage during execution
- Flash Memory: Reprogrammable memory for storing code/data

Input/output (I/O) Devices:

- **Input Devices:** Sensors, switches, keypads, microphones
- **Output Devices:** LEDs, displays, buzzers, motors, actuators
- Provide interaction between the embedded system and the external world.

Timers and Counters:

- Generate precise time delays
- Count events or pulses
- Essential for real-time control (e.g., motor control, data sampling)

Communication Interfaces:

- Allow data exchange with other systems or devices:
- UART (Serial Communication)
- SPI (Serial Peripheral Interface)
- I²C (Inter-Integrated Circuit)
- USB, CAN, Ethernet

Power Supply:

Converts AC to DC and provides stable voltage to the system.

Includes:

- Voltage regulators
- Battery backup
- Power management circuits

Example-Microcontroller:

Device	Main Controller	Function
Arduino UNO	ATmega328P	DIY electronics, robotics
Raspberry Pi	ARM Cortex-A	IoT, multimedia
ESP32	Xtensa Dual-Core	Wi-Fi & Bluetooth IoT